Physical chemistry cumulative exam

October 20, 2006

1. The specific (per kilogram) heat capacity of iron is C_P=0.450 kJ K^{-1} kg^{-1}; the specific heat capacity of water is C_P=4.184 kJ K^{-1} kg^{-1}. The gas constant is R=8.3145 J mol^{-1} K^{-1}.

A hot iron block of 2 kg mass is dropped into 1 L room-temperature (20 ^{o}C) water in a thermally insulated container.

(a) Determine the temperature of the system after thermal equilibrium has been reached. (15 p)

(b) Calculate the change of the enthalpy and entropy of the total system (iron block + water) (50 p)

(c) The atomic mass of iron is 55.85 g/mol. Calculate the molar heat capacity of iron. An empirical rule (Dulong-Petit rule) states that almost all heavy metals have a molar heat capacity close to that of iron. What is the theoretical basis of the Dulong-Petit rule? Calculate the ideal value of the molar heat capacity of iron from a fundamental constant. (25 p)

(d) Note that the heat capacity of water is very high compared to iron. Why has water such a high heat capacity? (10 p)